

The Bayesian Lasso

Park & Casella (2008), JASA 103(482), 681–686

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Presentation

The Lasso (Tibshirani, 1996)

Linear regression model

$$\mathbf{y} = \mu \mathbf{1}_n + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim N_n(0, \sigma^2 \mathbf{I}_n).$$

The Lasso estimator solves

$$\min_{\boldsymbol{\beta}} (\tilde{\mathbf{y}} - \mathbf{X}\boldsymbol{\beta})^\top (\tilde{\mathbf{y}} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^p |\beta_j|,$$

where $\tilde{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}_n$.

- Coefficient shrinkage (variance reduction)
- Sparse solutions (variable selection)

The L_1 penalty corresponds to a Laplace (double-exponential) prior:

$$\pi(\beta_j|\sigma^2) \propto \frac{\lambda}{\sqrt{\sigma^2}} \exp\left(-\frac{\lambda}{\sqrt{\sigma^2}}|\beta_j|\right).$$

The Lasso estimator is the posterior mode under independent Laplace priors on $\beta_1, \dots, \beta_p$.

A full Bayesian analysis offers:

- Credible intervals for all parameters (including σ^2)
- Uncertainty quantification
- Methods for choosing λ

Conditioning on σ^2 guarantees a unimodal full posterior

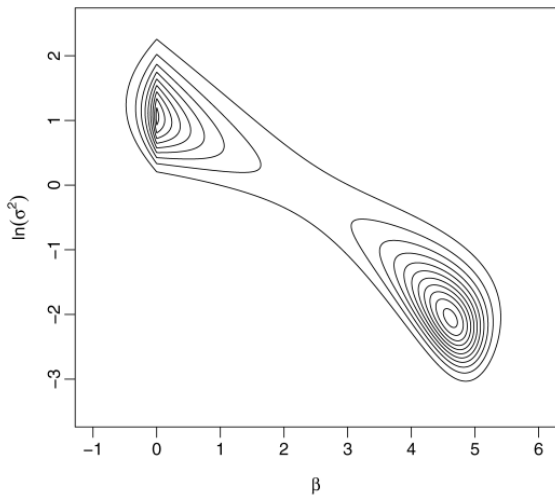


Figure B.1. Contour plot of an artificially generated posterior density of $(\beta, \ln(\sigma^2))$ of the form (B.2) that manifests bimodality. The logarithm of σ^2 is used only because it provides better visual scaling.

Conditional Laplace prior:

$$\pi(\boldsymbol{\beta} \mid \sigma^2) = \prod_{j=1}^p \frac{\lambda}{2\sqrt{\sigma^2}} \exp\left(-\frac{\lambda|\beta_j|}{\sqrt{\sigma^2}}\right), \quad \pi(\sigma^2) = 1/\sigma^2.$$

Conditional on σ^2 :

- Guarantees a unimodal joint posterior (Appendix A)
- Without conditioning, posterior can be bimodal (Appendix B)
- Bimodality $\Rightarrow$ slow Gibbs convergence, less meaningful point estimates

Scale Mixture of Normals (Andrews & Mallows, 1974)

The Laplace distribution admits the representation:

$$\frac{a}{2} e^{-a|z|} = \int_0^\infty \frac{1}{\sqrt{2\pi s}} e^{-z^2/(2s)} \cdot \frac{a^2}{2} e^{-a^2 s/2} ds, \quad a > 0.$$

A Laplace prior on β_j is equivalent to:

$$\beta_j \mid \tau_j^2 \sim N(0, \tau_j^2), \quad \tau_j^2 \sim \text{Exp}(\lambda^2/2).$$

This converts a non-conjugate problem into a conditionally conjugate one, enabling Gibbs sampling.

The Full Hierarchical Model

$$\begin{aligned}\mathbf{y} \mid \mu, \mathbf{X}, \boldsymbol{\beta}, \sigma^2 &\sim N_n(\mu \mathbf{1}_n + \mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n) \\ \boldsymbol{\beta} \mid \sigma^2, \tau_1^2, \dots, \tau_p^2 &\sim N_p(0_p, \sigma^2 \mathbf{D}_\tau), \quad \mathbf{D}_\tau = \text{diag}(\tau_1^2, \dots, \tau_p^2) \\ \tau_1^2, \dots, \tau_p^2 &\stackrel{\text{iid}}{\sim} \text{Exp}(\lambda^2/2) \\ \sigma^2 &\sim 1/\sigma^2 \quad (\text{or inverse-gamma})\end{aligned}$$

- μ given an independent flat prior, then marginalized (it is rarely of interest)
- Marginalizing $\tau_1^2, \dots, \tau_p^2$ recovers the conditional Laplace prior

Full Conditional Distributions

All full conditionals are tractable:

β (block update)

$$\beta \mid \text{rest} \sim N_p(\mathbf{A}^{-1} \mathbf{X}^\top \tilde{\mathbf{y}}, \sigma^2 \mathbf{A}^{-1}), \quad \mathbf{A} = \mathbf{X}^\top \mathbf{X} + \mathbf{D}_\tau^{-1}$$

σ^2

Inverse-Gamma with shape $(n-1)/2 + p/2$ and scale $\frac{1}{2}(\tilde{\mathbf{y}} - \mathbf{X}\beta)^\top (\tilde{\mathbf{y}} - \mathbf{X}\beta) + \frac{1}{2}\beta^\top \mathbf{D}_\tau^{-1} \beta$

$\tau_1^2, \dots, \tau_p^2$ (conditionally independent)

$1/\tau_j^2 \mid \text{rest} \sim \text{Inverse-Gaussian}(\mu', \lambda')$ with

$$\mu' = \sqrt{\lambda^2 \sigma^2 / \beta_j^2}, \quad \lambda' = \lambda^2$$

Empirical Bayes via Monte Carlo EM

Casella (2001) does maximum marginal likelihood: treat β, σ^2, τ^2 as “missing data”, iterate EM with Gibbs-based E-steps.

At iteration k :

$$\lambda^{(k)} = \sqrt{\frac{2p}{\sum_{j=1}^p \mathbb{E}_{\lambda^{(k-1)}}[\tau_j^2 \mid \tilde{\mathbf{y}}]}}$$

where the expectation is approximated by the Gibbs sample average.

Initialization (works well in practice):

$$\lambda^{(0)} = \frac{p \sqrt{\hat{\sigma}_{\text{LS}}^2}}{\sum_{j=1}^p |\hat{\beta}_j^{\text{LS}}|}$$

Estimates eventually drift around the true MLE.

Fully Bayesian: Hyperprior on λ^2

Class of conjugate gamma hyperpriors on λ^2 (not λ):

$$\pi(\lambda^2) = \frac{\delta^r}{\Gamma(r)} (\lambda^2)^{r-1} e^{-\delta \lambda^2}, \quad r, \delta > 0.$$

Resulting full conditional:

$$\lambda^2 \mid \text{rest} \sim \text{Gamma}\left(p + r, \frac{1}{2} \sum_{j=1}^p \tau_j^2 + \delta\right).$$

- Avoid $1/\lambda^2$ (improper posterior)
- Choose (r, δ) so the hyperprior is diffuse but concentrated near the MLE
- λ^2 simply joins the other parameters in the Gibbs sampler

Diabetes Data (Efron et al., 2004; $n = 442$, $p = 10$)

Empirical findings:

- Bayesian Lasso paths appear as a compromise between Lasso and ridge
- Smooth like ridge, but shape mimics Lasso
- Pulls weak coefficients toward zero faster than ridge
- Vertical lines: n -fold CV (Lasso) vs. marginal MLE (Bayesian Lasso)

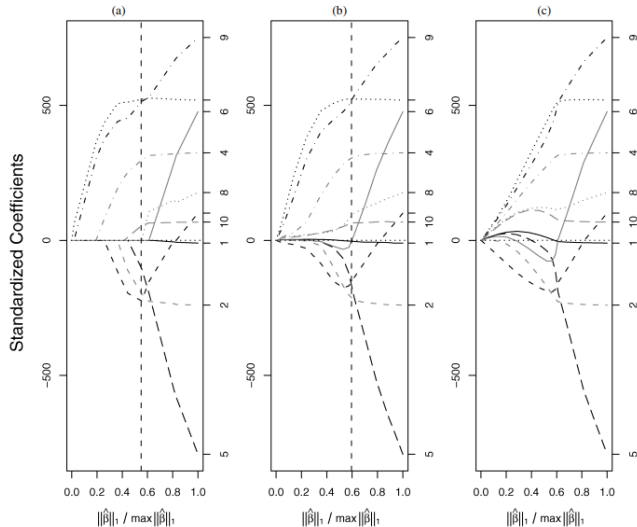


Figure 1. Lasso (a), Bayesian Lasso (b), and ridge regression (c) trace plots for estimates of the diabetes data regression parameters versus the relative L_1 norm, with vertical lines for the Lasso and Bayesian Lasso indicating the estimates chosen by n -fold cross-validation and marginal maximum likelihood. The Bayesian Lasso estimates were posterior medians computed over a grid of λ values, using 10,000 consecutive iterations of the Gibbs sampler of Section 2 (after 1,000 burn-in iterations) for each λ .

Comparison with Lasso Estimates

Posterior medians with 95% credible intervals (Figure 2):

- Bayesian posterior medians remarkably close to Lasso estimates
- All Lasso estimates lie inside the credible intervals
- 4 least-squares estimates lie outside the credible intervals
- Native uncertainty quantification

Bayesian Lasso estimates are not exactly zero. The Bayesian Lasso provides shrinkage and uncertainty, but not selection, unless paired with a thresholding rule on the credible intervals.

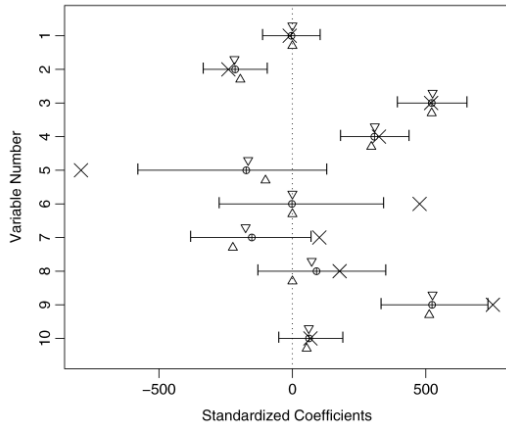


Figure 2. Posterior median Bayesian Lasso estimates ($\oplus$) and corresponding 95% credible intervals (equal-tailed) with λ selected according to marginal maximum likelihood (Sec. 3.1). Overlaid are the least squares estimates ($\times$), Lasso estimates based on n -fold cross-validation (Δ), and Lasso estimates chosen to match the L_1 norm of the Bayes estimates (∇). The variables were described by Efron et al. (2004): (1) age, (2) sex, (3) bmi, (4) map, (5) tc, (6) ldl, (7) hdl, (8) tch, (9) ltg, and (10) glu.

Extensions of the Framework

The scale-mixture hierarchy is modular: change the prior on τ_j^2 to mimic other procedures.

Bridge regression

Penalty $\lambda \sum_j |\beta_j|^q$, $0 < q \leq 2$.

Representable as scale mixture using a positive stable mixing density (West, 1987; Gneiting, 1997). Gibbs sampler less clean.

“Huberized Lasso” (robust)

Replace σ^2 by observation-specific σ_i^2 with a generalized inverse-Gaussian prior on (σ_i^2, ρ^2) .
Yields a hyperbolic-loss likelihood: quadratic near zero, linear in tails.

Generalized linear models also accessible (e.g. Bae & Mallick, 2004 for probit).

The Bayesian Lasso provides:

- A clean hierarchical model for L_1 -penalized regression
- A simple, efficient Gibbs sampler (closed-form full conditionals)
- Credible intervals for all parameters, including σ^2
- Two principled methods for choosing λ : EM and hyperprior
- A modular framework, extensible to bridge regression, robust loss, GLMs

Limitations:

- Laplace tails are exponential
- No pole at zero $\Rightarrow$ does not aggressively shrink noise
- No global sparsity parameter

$\rightarrow$ Motivates the horseshoe prior (Carvalho, Polson, Scott, 2010).

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Thank you!

Questions?